

PRD_Tasks_Module

Tasks Module and Task Page

Full Product Requirements Document for Product, UX, Frontend, Backend, QA and DevOps Teams

The Tasks Module is the platform's central collaboration and work-orchestration layer. It connects selected users around specific work and links tasks to projects, units, clients, documents, meetings, approvals, finance records and other business objects.

Current access principle: all authenticated users may access the complete task feature set during this implementation phase. Detailed role-based restrictions will be defined in a later permissions PRD.

Status: Confirmed design and functional scope
Version: 1.0
Output: Developer-ready specification

1. Document Control

Field	Value
Document	PRD_Tasks_Module - Tasks Module and Task Page
Audience	Product Manager, UI/UX, Frontend, Backend, QA, DevOps, Security
Primary pages	Tasks Module Page, Task Page
Supporting flows	Create Task, Participant Selector, Checklist, Approval Request, Document Upload, Dependency Setup, Recurrence Setup, Saved Views, Activity Timeline
Permissions	All authenticated users enabled initially; restrictions deferred to a later PRD

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3. Product Vision and Goals

The Tasks Module must become the common operating layer through which users coordinate work across the entire platform.

- Connect explicitly selected users around one piece of work.
- Allow every task to carry context, documents, comments, approvals, checklist items, dependencies and decisions.
- Keep collaboration private to selected participants instead of broadcasting work to an entire department.
- Make tasks usable from any major module without duplicating business data.
- Support both simple To-Do work and complex multi-step work packages.
- Provide a premium, fast and responsive user experience consistent with the Projects and Project Page design system.

A task is not only a reminder. It is a controlled collaborative record of responsibility, communication, attached evidence, approvals and outcome.

4. Core Principles

- Participant-based privacy: tasks are visible to the creator, selected participants, watchers and approvers.
- Explicit collaboration: users are added intentionally; departments do not automatically gain access.
- Single source of truth: linked documents, projects, units and clients remain owned by their original modules.
- Personal organization: Star, saved views and layout preferences are stored per user.
- Traceability: meaningful changes appear in the Task Activity Timeline.
- Responsive parity: desktop and mobile support the same core actions.
- Premium simplicity: dense data remains readable through hierarchy, spacing and restrained animation.

5. Information Architecture

Primary navigation: Tasks Module Page -> Task Page

Supporting flows: Create Task -> Participant Selection -> Checklist/Subtasks -> Documents -> Approval -> Activity

Linked context: Project, Structure, Unit, Client, Document, Meeting, Approval, Work Package, Contract, Finance Record, Government & Legal Record, Employee, Company and Department metadata.

- Tasks Module Page: global task discovery and personal work management.
- Task Page: full workspace for one task.
- Quick Task Drawer: optional lightweight preview from tables or linked modules.
- Create Task Flow: full task setup.
- Task Participant Selector: user-only selection; department reference is metadata.
- Shared Task Checklist: collaborative To-Do items.
- Task Approval Request Flow: creates a linked Approval record.
- Task Activity Timeline: chronological audit-oriented event stream.

6. Tasks Module Page

The Tasks Module Page is the user's central task workspace across all projects and business records.

- Page title and description
- Create Task button

- Summary cards
- Task scope tabs
- Filters and sorting
- Layout switcher
- Saved views
- Task results area
- Pagination or infinite loading
- Optional task preview drawer

Summary cards may show My Tasks, Starred, Due Today, Upcoming, Overdue, Completed and Awaiting My Approval. Clicking a summary card applies the corresponding task scope.

TASK-MOD-001 - Provide List, Board/Kanban, Calendar and Timeline layouts.

TASK-MOD-002 - Persist the last selected layout per user.

TASK-MOD-003 - Keep filters and scope consistent when switching layouts.

TASK-MOD-004 - Support personal saved views with layout, filters, sorting, grouping and visible fields.

TASK-MOD-005 - Allow each user to Star tasks privately.

TASK-MOD-006 - Provide task scopes: My Tasks, Created by Me, Assigned to Me, Participating, Starred, Mentioned, Following, Awaiting My Approval, Overdue, Upcoming, Completed and All Accessible Tasks.

Task row/card fields

- Title and Task ID
- Linked project or record
- Status
- Official priority
- Participants
- Responsible user
- Start and due date
- Checklist progress
- Approval status
- Attachment count
- Comment count
- Personal Star
- Last update
- Overdue or blocked indicators

Tasks Module actions

- Create Task
- Duplicate Task
- Archive Task
- Bulk assign participants
- Bulk change status
- Bulk change priority
- Bulk change due date
- Export tasks
- Create from template
- Save current view

7. Task Page

The Task Page is the full collaborative workspace for one task.

Header requirements

- Personal Star
- Task title and Task ID
- Status selector
- Priority selector
- Progress
- Due date
- Participants
- Edit action
- Mark Complete action
- More actions menu

Core tabs or anchored sections

- Overview
- Checklist
- Subtasks
- Comments
- Documents
- Approvals
- Activity

Right-side information panels on desktop may include Task Details, Linked Records, Dependencies, Approval and Actions. On mobile, these become collapsible sections or a More sheet.

TASK-PAGE-001 - Display creator, responsible user, participants and watchers separately.

TASK-PAGE-002 - Allow participants to add or remove checklist items and update task content.

TASK-PAGE-003 - Show linked records as clickable cards.

TASK-PAGE-004 - Provide a permanent activity timeline.

TASK-PAGE-005 - Maintain feature parity between desktop and mobile.

8. Task Creation and Assignment

- Title
- Detailed description
- Creator
- Responsible user
- Selected participants
- Watchers
- Linked records
- Status
- Priority
- Start date
- Due date

- Reminder
- Checklist
- Attachments
- Recurrence
- Dependencies
- Approval requirement

Tasks are normally shared between 2-4 explicitly selected users. The system must not automatically add an entire department.

9. Shared Checklist and Subtasks

- Checklist item title
- Optional participant owner
- Optional due date
- Notes
- Attachment
- Completion status
- Created by
- Completed by
- Completion timestamp

Every check/uncheck action must appear in the Task Activity Timeline with user and timestamp.

10. Comments, Threads and Mentions

- Plain and rich text comments
- Threaded replies
- @user mentions
- Emoji reactions
- File attachments
- Linked platform documents
- Image previews
- Convert comment to checklist item
- Convert comment to subtask
- Pin important comment
- Record formal decision

Mentioning a user does not automatically expose the task. If the mentioned person is not a participant, the UI must prompt the author to add them.

11. Documents and Attachments

- Upload files directly to the task
- Link existing platform documents
- Preview supported files
- Download attachments
- Open source Document Page

- Mark attachment as deliverable
- Request document review
- Request document approval
- Upload document revision where supported
- Show uploader and upload timestamp

Files should be stored through the platform's central document/storage service. The task stores a relation, not an isolated duplicate.

12. Approval Integration

- Approval title
- Decision required
- Approver
- Supporting documents
- Description
- Deadline
- Cost impact
- Timeline impact
- Technical impact
- Related platform records

The Approval record remains owned by the Approvals Module and is linked back to the task.

13. Linked Records

- Project
- Building / Structure / Phase / Zone
- Floor / Level
- Unit
- Client
- Document
- Meeting
- Approval
- In-Progress Work
- Contract
- Finance record
- Government & Legal record
- Employee
- Company
- Department as metadata

One task may link to multiple records. Primary links should be prominent; secondary links may remain compact.

14. Status, Priority and Star Logic

Recommended task status lifecycle

- Draft

- To Do
- In Progress
- Waiting
- Blocked
- Under Review
- Completed
- Cancelled
- Archived

Official priority

- Low
- Normal
- High
- Critical

Personal Star

- Private per user
- Does not change official priority
- Does not notify other users
- Adds task to Starred Tasks
- Can be toggled from Module Page and Task Page

15. Views, Filters and Saved Views

List View

- Dense operational table
- Column customization
- Multi-sort
- Bulk actions
- Inline status changes

Board View

- Columns by status or custom grouping
- Drag-and-drop status change
- Compact task cards
- Swimlanes optional

Calendar View

- Tasks by due date
- Drag to reschedule
- Month/week/day modes
- Overdue emphasis

Timeline View

- Start-to-due range
- Dependencies
- Milestones
- Zoom controls

Core filters

- Project
- Company
- Unit
- Client
- Participant
- Creator
- Responsible user
- Status
- Priority
- Due date
- Overdue
- Approval status
- Has attachments
- Incomplete checklist
- Starred by me
- Updated recently

Saved view data

- Name
- Owner
- Layout
- Scope
- Filters
- Sort
- Grouping
- Visible fields
- Optional sharing later

16. Dependencies and Recurrence

Dependency types

- Blocked by
- Blocking
- Related to
- Parent of subtask
- Dependent on approval
- Dependent on document
- Dependent on meeting decision

Recurring task options

- Daily
- Weekly
- Monthly
- Custom schedule

- Until date
- Occurrence count

Each generated occurrence must remain traceable to the recurrence series and retain its own completion and activity data.

17. Notifications and Escalations

- Added as participant
- Assigned responsibility
- Mentioned
- New comment
- Checklist update
- Document uploaded
- Approval requested
- Approval result
- Status changed
- Due date changed
- Upcoming deadline
- Task overdue
- Task completed
- Task reopened
- Dependency unblocked

Users should be able to configure notification preferences by event type and channel. Department-wide notification is not the default.

18. Activity Timeline and Audit

- Task created
- Participant added or removed
- Responsible user changed
- Status changed
- Priority changed
- Due date changed
- Checklist item created, checked or unchecked
- Subtask created or completed
- Comment added
- Document uploaded or linked
- Approval requested or completed
- Dependency added or removed
- Task completed, reopened, cancelled or archived

Activity entries must include acting user, timestamp, event type, readable description and relevant record identifiers.

19. Responsive Design

Desktop

- Persistent left navigation
- Wide task list/table
- Optional preview drawer
- Task Page multi-column layout
- Right-side details panels

Tablet

- Collapsible sidebar
- Two-column Task Page
- Scrollable tabs
- Details panel becomes drawer

Mobile

- Single-column content
- Sticky compact header
- Scrollable section tabs
- Participants as compact avatars
- Checklist placed early
- Bottom action bar for Comment, Attach, Subtask and Complete
- More sheet for secondary actions

All core actions must remain available on mobile. Responsive design must not reduce the module to read-only mode.

20. Premium Motion and Micro-interactions

- Task row hover: 120-180 ms subtle elevation, warm border tint and soft background transition.
- Summary card hover: slight lift of 2-4 px, shadow increase and icon scale to 1.03.
- Star toggle: 160 ms scale and fade with restrained gold highlight.
- Status chip: smooth color transition and short dropdown reveal.
- Board drag: card lift, shadow and placeholder animation.
- Checklist completion: checkmark draw, text fade/strike and progress bar interpolation.
- Task preview drawer: 220-280 ms slide with opacity transition.
- Tab switching: underline glide and content crossfade.
- Comment posting: optimistic insertion with subtle fade/slide.
- Upload: progress ring/bar and completion confirmation.
- Mobile bottom sheet: restrained spring-like transition.
- Loading: skeleton placeholders matching final geometry.

All animations must respect reduced-motion accessibility settings.

21. Data Model

Entity	Purpose	Key fields
Task	Primary work record	id, tenant_id, company_id, title, description, status, priority, creator_id, responsible_user_id, start_at, due_at, completed_at, archived_at
TaskParticipant	Explicit task access	task_id, user_id, role, added_by, added_at
TaskWatcher	Follower relation	task_id, user_id, created_at

Entity	Purpose	Key fields
TaskStar	Private per-user star	task_id, user_id, created_at
TaskChecklistItem	Shared To-Do	task_id, title, owner_id, due_at, completed, completed_by, completed_at, sort_order
TaskSubtask	Child task relation	parent_task_id, child_task_id, sort_order
TaskComment	Discussion	task_id, author_id, body, parent_comment_id, edited_at, deleted_at
TaskAttachment	File/document relation	task_id, file_id/document_id, uploaded_by, attachment_type
TaskLink	Generic linked record	task_id, entity_type, entity_id, is_primary
TaskDependency	Dependency graph	task_id, depends_on_task_id, type
TaskRecurrence	Recurring schedule	task_id, rule, timezone, until_at, count
TaskActivity	Timeline event	task_id, actor_id, event_type, payload_json, created_at
TaskSavedView	Personal saved view	user_id, name, layout, scope, filters_json, sort_json, columns_json

22. API Requirements

Suggested REST endpoints. Equivalent GraphQL operations are acceptable if the platform standard uses GraphQL.

- GET /tasks
- POST /tasks
- GET /tasks/{id}
- PATCH /tasks/{id}
- DELETE /tasks/{id} or archive
- POST /tasks/{id}/participants
- DELETE /tasks/{id}/participants/{userId}
- POST /tasks/{id}/star
- DELETE /tasks/{id}/star
- POST /tasks/{id}/checklist
- PATCH /tasks/{id}/checklist/{itemId}
- POST /tasks/{id}/comments
- POST /tasks/{id}/attachments
- POST /tasks/{id}/links
- POST /tasks/{id}/dependencies
- POST /tasks/{id}/approval-requests
- GET /tasks/{id}/activity
- GET /task-views
- POST /task-views

All mutation endpoints must validate tenant scope, task participation and optimistic concurrency where required.

23. Frontend Architecture

- TasksModulePage
- TaskSummaryCards
- TaskScopeTabs

- TaskFilterBar
- TaskLayoutSwitcher
- TaskListTable
- TaskBoard
- TaskCalendar
- TaskTimeline
- TaskPreviewDrawer
- TaskPage
- TaskHeader
- TaskParticipants
- TaskChecklist
- TaskSubtasks
- TaskComments
- TaskDocuments
- TaskApprovals
- TaskActivityTimeline
- TaskDetailsPanel
- LinkedRecordsPanel
- DependenciesPanel
- CreateTaskModal/Page

Recommended client behavior

- Server state caching
- Optimistic updates for Star, checklist, comments and status
- Virtualized rows for large task lists
- URL-based scope, filters and layout
- Debounced search
- Accessible keyboard navigation
- Reusable design-system components

24. Backend and Background Jobs

- Task CRUD service
- Participation and access service
- Checklist service
- Comment and mention parser
- Attachment/document linking service
- Approval integration service
- Dependency graph validation
- Recurrence scheduler
- Reminder and overdue scheduler
- Notification dispatch
- Search indexing
- Activity event writer
- Export generation

Background jobs must be idempotent. Recurring task creation, reminders and overdue notifications must not generate duplicates when retried.

25. Search and Performance

- Search by title, Task ID, description, project, client, participant and linked record.
- Index task title, description, IDs and selected linked-record metadata.
- Paginate task lists.
- Use cursor pagination for very large datasets.
- Use virtualized rendering for large lists.
- Lazy-load comments, activity and attachments where necessary.
- Target interactive response below 200 ms for cached UI actions.
- Use optimistic UI for low-risk mutations.

26. Validation and Error Handling

- Task title required
- Due date cannot precede start date
- Prevent circular dependencies
- Prevent self-dependent tasks
- Validate recurrence rule
- Validate attachment size and type
- Prevent completion when required approval remains unresolved, if configured
- Show conflict message when concurrent edits occur
- Allow retry for failed uploads and comments
- Preserve unsaved draft text where possible

28. Analytics and Reporting

- Open tasks
- Completed tasks
- Overdue tasks
- Tasks by status
- Tasks by priority
- Completion time
- Checklist completion rate
- Workload by participant
- Approval waiting time
- Blocked task duration
- Tasks by linked project or company
- Personal Starred count

27. Security and Access Model

Current implementation phase

- All authenticated users can create and use tasks.
- Access to a specific task is participant-based.

- Creator, participants, watchers and approvers can access the task.
- Departments do not automatically inherit access.
- Linked record access must not be widened by the task.

Future permissions PRD

- Role-based creation and deletion rules
- Commercial and financial data visibility
- External participant restrictions
- Company/group boundaries
- Administrative override and audit

29. Testing Strategy

Unit tests

- Status transitions
- Star privacy
- Checklist completion
- Dependency validation
- Recurrence generation
- Participant access

Integration tests

- Task creation with linked record
- Approval request from task
- Document upload
- Notification dispatch
- Search indexing

End-to-end tests

- Create collaborative task
- Add participant
- Complete checklist
- Post comment
- Upload file
- Request approval
- Complete task
- Switch List/Board/Calendar/Timeline
- Save personal view
- Responsive mobile actions

Visual regression

- Desktop Tasks Module
- Desktop Task Page
- Tablet layout
- Mobile Task Page
- Reduced-motion mode

30. Acceptance Criteria

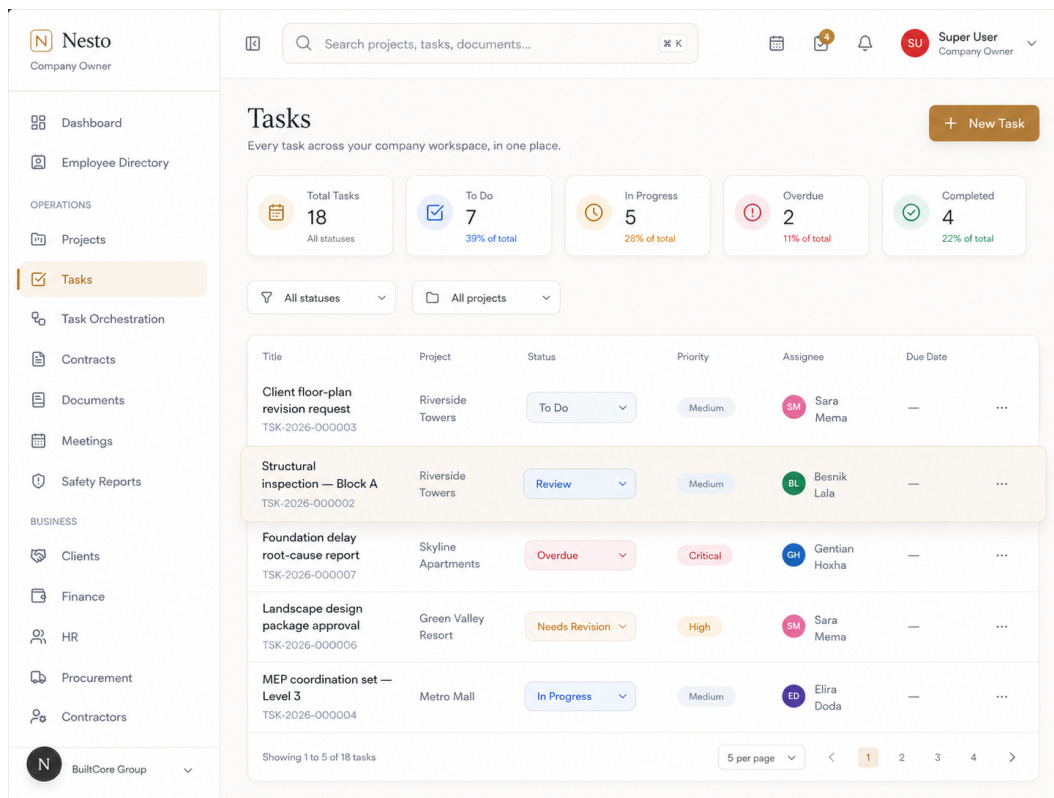
- AC-01 - Users can create tasks and explicitly select participants.
- AC-02 - Tasks are not automatically visible to an entire department.
- AC-03 - Each user can privately Star a task.
- AC-04 - Users can switch between List, Board, Calendar and Timeline.
- AC-05 - Selected layout persists per user.
- AC-06 - Saved views preserve layout, filters, sort and visible fields.
- AC-07 - Any participant can check or uncheck shared checklist items.
- AC-08 - Checklist actions appear in the activity timeline.
- AC-09 - Task Page supports comments, mentions, documents, subtasks and approvals.
- AC-10 - Tasks can link to all major platform records.
- AC-11 - Approval requests can be created from the Task Page.
- AC-12 - Desktop and mobile support all core actions.
- AC-13 - Premium hover states and micro-interactions are implemented.
- AC-14 - Reduced-motion preferences are respected.
- AC-15 - Large task lists remain performant.

31. Page and Flow Registry

- Tasks Module Page
- Task Page
- Create Task
- Task Participants Selector
- Shared Task Checklist
- Starred Tasks
- Task Assignment Flow
- Task Approval Request Flow
- Task Document Upload
- Task Dependency Setup
- Task Recurrence Setup
- Task Activity Timeline
- Saved Task Views
- Task Preview Drawer
- Task Template
- Duplicate Task Flow
- Archive Task Flow

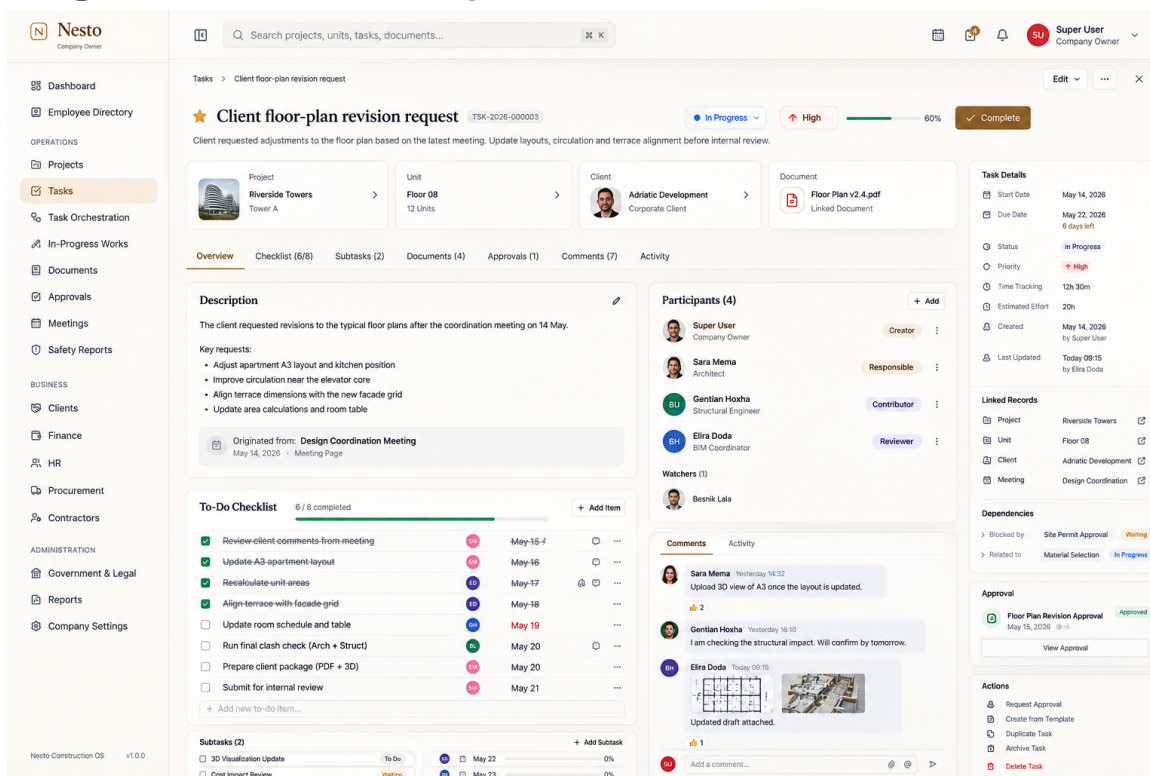
32. Visual References

Tasks Module - Confirmed Desktop Direction



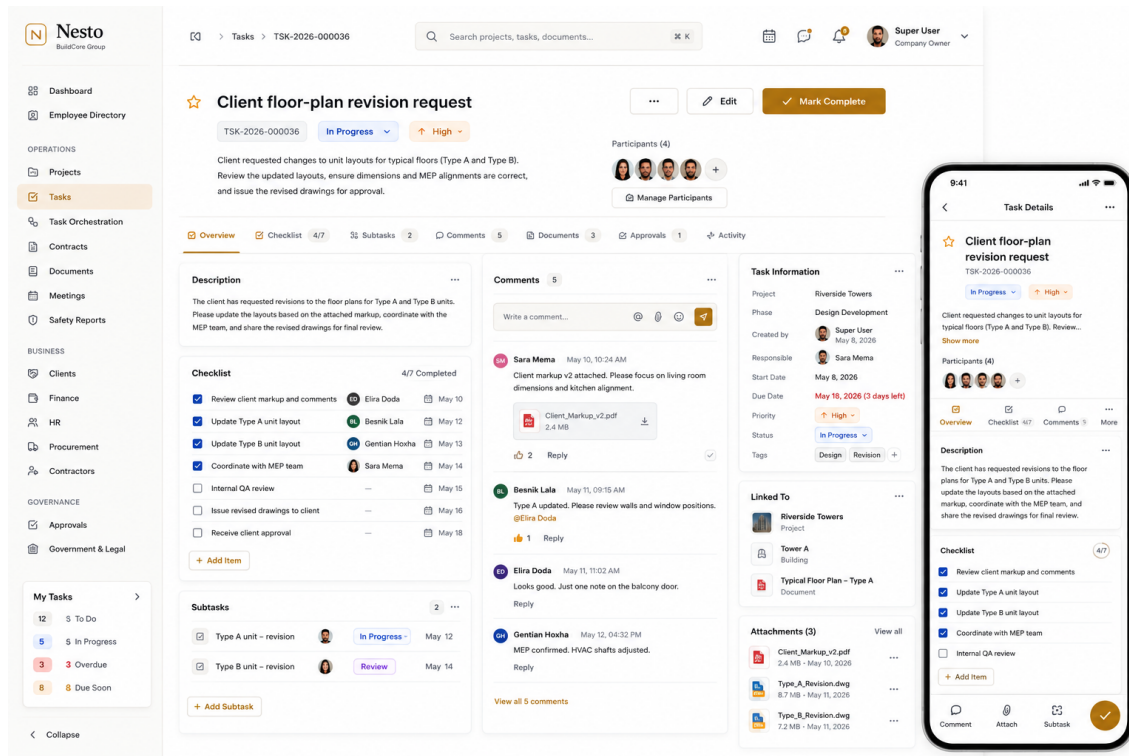
Reference image: Tasks Module - Confirmed Desktop Direction.

Task Page - Confirmed Desktop Direction



Reference image: Task Page - Confirmed Desktop Direction.

Task Page - Responsive Desktop and Mobile Direction



Reference image: Task Page - Responsive Desktop and Mobile Direction.